



M.A.M. SCHOOL OF ENGINEERING

(An Autonomous Institution)

Accredited by NAAC | Approved by AICTE-New Delhi | Affiliated to Anna University-Chennai
Trichy-Chennai Trunk Road, Siruganur, Tiruchirappalli -621 105



Department of Electrical and Electronics Engineering Academic

Year 2025-2026 (Even Semester)

Unit - I

Sub.Code/Sub.Name : EE3601-Protection & Switch Gear

Date :

Year/Sem. : III / VI

Duration : 3 Hrs

Max.Marks : 100

	PART A	M.A	CO	BTL
1.	What are the different types of faults in power system?	2	CO1	BT1
2.	What is earthing?	2	CO1	BT1
3.	What is backup protection?	2	CO1	BT2
4.	State any four functions of protective relay.	2	CO1	BT1
5.	What is the difference between primary and backup protection?	2	CO1	BT1
6.	What is protective zone?	2	CO1	BT1
7.	What is time setting multiplier?	2	CO1	BT2
8.	List the basic requirements or essential qualities of protective relaying.	2	CO1	BT1
9.	What is dead spot or blind spot in a power system?	2	CO1	BT1
10.	Why the secondary of CT should not be open?	2	CO1	BT2

	Part B	M.A	CO	BTL
11.	Describe the essential qualities of a protective relaying.	16	CO1	BT2
12.	Discuss in detail about different protective schemes.	16	CO1	BT2
13.	Explain Arc suppression coil (Peterson coil) earthing with neat diagram.	16	CO1	BT2
14.	What do you understand by a zone of protection? Discuss various zones of protection with neat diagram.	16	CO1	BT4
15.	Enumerate the basic ideas of insulation coordination.	16	CO1	BT4



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Department of Electrical and Electronics Engineering

Academic Year 2025-2026 (Even Semester)

Unit - II

Sub.Code/Sub.Name : EE3601-Protection & Switch Gear

Date :

Year/Sem. : III / VI

Duration : 3 Hrs

Max.Marks : 100

	PART A	M.A	CO	BTL
1.	Compare static and electromagnetic relays.	2	CO2	BT2
2.	What are over current and under current relay?	2	CO2	BT2
3.	Compare the difference between directional over current relay and non-directional over current relay.	2	CO2	BT2
4.	Give the application of electromagnetic relays.	2	CO2	BT1
5.	Write the function of under frequency relay.	2	CO2	BT2
6.	Why under frequency relay is required in a power system?	2	CO2	BT2
7.	Show the characteristics of an impedance relay on R-X diagram.	2	CO2	BT3
8.	Give the principle of negative sequence relay.	2	CO2	BT2
9.	Write the torque equation OF THE UNIVERSALRELAY.		CO2	BT2
10.	A relay is connected to 400/5 ratio current transformer with current setting of 150%. calculate the PSM when the fault current is 4000A.	2	CO2	BT3

	Part B	M.A	CO	BTL
11.	Describe the construction and operation of an inductive type directional over current relay with neat diagram also explain its operational characteristics.	13	CO2	BT4
12.	Explain an impedance type distance relay showing its characteristics on R-X Diagram and reactance relay showing its characteristics on R-X diagram.	13	CO2	BT2
13.	With neat diagram explain the various types of electromagnetic relays.	13	CO2	BT3
14.	Describe the construction and principle of operation of non-directional induction type over current relay.	13	CO2	BT4
15.	Derive the universal torque equation and also explain negative sequence relay.	13	CO2	BT4

Faculty signature

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Department of Electrical and Electronics Engineering Academic

Year 2025-2026 (Even Semester)

Unit - III

Sub.Code/Sub.Name : EE3601-PROTECTION AND SWITCH GEAR

Date :

Year/Sem. : III / VI

Duration : 3 Hrs

Max.Marks : 100

	PART A	M.A	CO	BTL
1.	Explain the secondary of CT should not be open.	2	CO3	K2
2.	Define the term burden on CT.	2	CO3	K1
3.	Show the applications of Buchholz's relay.	2	CO3	K3
4.	Discuss the shortcomings of differential protection scheme as applied to power transformer.	2	CO3	K4
5.	Examine over fluxing protection of a transformer?	2	CO3	K4
6.	List the common faults that occur in a generator.	2	CO3	K1
7.	Infer the causes of over speed and how alternators are protected from it?	2	CO3	K1
8.	Classify the types of bus bar protection.	2	CO3	K3
9.	Explain the faults associated with a transformer?	2	CO3	K2
10.	Explain time-graded system protection?	2	CO3	K2

	Part B	M.A	CO	BTL
11.	Describe the differential protective scheme of transformer.	16	CO3	K3
12.	Explain the principle of percentage biased differential protection with necessary diagrams. Also discuss its applications	16	CO3	K2
13.	Describe the differential pilot wire method of protection of feeder	16	CO3	K2
14.	Justify the ways of protection in AC motor.	16	CO3	K4
15.	Describe the types of protective schemes employed for the protection of Busbar.	16	CO3	K3



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Department of Electrical & Electronics Engineering Academic Year 2025-2026 (Even Semester)

Unit - II

Sub.Code/Sub.Name : EE3601-PROTECTION AND SWITCH GEAR

Date :

Year/Sem. : III / VI

Duration : 3 Hrs

Max.Marks : 100

	PART A	M.A	CO	BTL
1.	Why under frequency relay is required in a power system?	2	CO2	K2
2.	Show the characteristics of an impedance relay on R-X diagram.	2	CO2	K3
3.	Give the principle of negative sequence relay.	2	CO2	K2
4.	Write the torque equation of the universal relay	2	CO2	K1
5.	A relay is connected to 400/5 ratio current transformer with current setting of 150%. calculate the PSM when the fault current is 4000A.	2	CO2	K3

	Part B	M.A	CO	BTL
11.	Describe the construction and principle of operation of non-directional induction type over current relay.	13	CO2	K2
12.	Derive the universal torque equation and also explain negative sequence relay.	13	CO2	K3
13.	With neat diagram explain the various types of electromagnetic relays.	13	CO2	K2

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Department of Electrical and Electronics Engineering
Academic Year 2025-2026 (Even Semester)

Unit - IV

CIA – III Exam

Sub.Code/Sub.Name : EE3601-protection and switch gear
Year/Sem. : III/VI

Date :
Duration : 3hrs
Max.Marks : 100

Part A (10 X 2 = 20 Marks) Answer All the Questions		M.A	CO	BTL
1.	Define static relay.	2	CO4	BTL1
2.	What are the merits of static relay.?	2	CO4	BTL1
3.	What are the types of comparators employed in static relay?	2	CO4	BTL1
4.	List the advantages of digital relay.	2	CO4	BTL1
5.	Compare numerical relay with electromechanical relay.	2	CO4	BTL2
6.	Draw the block diagram of numerical over current relay.	2	CO4	BTL2
7.	What are the different components of the digital relay?	2	CO4	BTL1
8.	Draw the distance relay characteristic curve.	2	CO4	BTL2
9.	Draw the multi-channel over current protection relay.	2	CO4	BTL2
10.	What are the characteristics of numerical relay?	2	CO4	BTL1

		Part B	M.A	CO	BTL
11.	a)	(i) Analyze the directional static overcurrent relay. (ii) explain the static distance relay.	8+8	CO4	BTL2
12.	a)	Explain the comparators in static relays and merits and demerits of static relay.	16	CO4	BTL2
13.	a)	Explain numerical over current protection of transmission lines.	16	CO4	BTL2
14.	a)	Explain the block diagram of numerical relay with necessary diagram.	16	CO4	BTL2
15.	a)	Explain numerical distance protection of transmission lines.	16	CO4	BT2

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Department of Electrical and Electronics Engineering
Academic Year 2025-2026(Even Semester)
CIA – III Exam

Unit - V

Sub.Code/Sub.Name : EE3601-protection and switch gear
Year/Sem. : III/VI

Date :
Duration : 3 Hrs
Max.Marks : 100

		Part A (5 X 2 = 10 Marks) Answer All the Questions			M.A	CO	BTL
1.		What is an arc?	2	CO5	BTL1		
2.		Mention the methods of arc interruption.	2	CO5	BTL1		
3.		Define restriking voltage.	2	CO5	BTL1		
4.		Define recovery voltage.	2	CO5	BTL1		
5.		Mention the different methods of high resistance arc interruption.	2	CO5	BTL1		
6.		What are the quenching factors in an OCB?	2	CO5	BTL1		
7.		What is the principle used in arc interruption in air-circuit breaker?	2	CO5	BTL1		
8.		How the arc is quenched in vacuum circuit breaker?	2	CO5	BTL1		
9.		What are the basic requirements of circuit breaker?	2	CO5	BTL1		
10.		What are the different types of oil circuit breaker?	2	CO5	BTL1		

		Part B			M.A	CO	BTL
11.	a	With a neat diagram explain the construction and working principle of (i) Air Blast circuit Breaker and (ii) vacuum circuit breaker.	8+8	CO5	BTL2		
12.	a	Write short notes on: (i) Current chopping (ii) Interruption of capacitive current	8+8	CO5	BTL2		
13.	a	Show the constructional layout of SF6 Breaker and give its advantages and disadvantages.	16	CO5	BTL2		
14.	a	With a neat sketch explain the principle of vacuum circuit breaker.	16	CO5	BTL2		

15	a	With a neat sketch explain the principle of (i) bulk oil circuit breaker and (ii) minimum oil circuit breaker.	8+8	CO5	BTL2
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